

No. of Functional Features Included in the Solution

Date	4 July 2026
Team ID	XXXXXX
Project Name	Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Preprocessing	Cleans and prepares the dataset for model training by selecting features and applying preprocessing techniques.	Data Preprocessing	Done	High
2	Feature Scaling	Standardizes numerical features using StandardScaler to improve model performance.	Machine Learning	Done	High
3	Machine Learning Model Training	Trains Decision Tree, Random Forest, KNN, and XGBoost models using historical flood data.	Machine Learning	Done	High
4	Model Evaluation	Evaluates trained models using accuracy, confusion matrix, and classification report to select the best model.	Machine Learning	Done	High
5	Flood Prediction	Predicts whether flood conditions are likely based on user-provided weather parameters.	Prediction Engine	Done	High
6	Web-Based User Interface	Provides an interface for entering weather data and displaying prediction results.	Frontend (HTML/CSS/JS)	Done	Medium
7	Flask Integration	Connects the frontend with the trained machine learning model and handles prediction requests.	Backend (Flask)	Done	High
8	Model Persistence	Saves and loads the trained model and scaler using Joblib for efficient predictions.	Model Storage	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Flood Prediction Form
2	Backend / Logic	4	Flask Routing, Prediction Engine, Model Evaluation, Input Processing
3	Database / Storage	1	Saved Model (floods.save), Scaler (transform.save), Dataset
4	API / Integration	1	Flask Integration between Frontend and Machine Learning Mode
5	Security / Authentication	0	Not Applicable (No Login or User Authentication)